

Activity14_Bryan

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#Armed Forces Data Wrangling

```
library(tidyr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tibble)
```

```
#Wrangle Data
```

```
col_names <- c("pay_grade", "army_male", "army_female", "army_total", "navy_male", "navy_female", "navy_to
Armed_forces <- read.csv("US_Armed_Forces_(6_2025) - Sheet1.csv")
colnames(Armed_forces) <- col_names
Armed_forces <- na.omit(Armed_forces)
col_names_keep <- c("pay_grade", "army_male", "army_female", "navy_male", "navy_female", "marinecorps_male"

Armed_forces <- Armed_forces %>%
  select(col_names_keep)
```

```
## Warning: Using an external vector in selections was deprecated in tidysselect 1.1.0.
## i Please use 'all_of()' or 'any_of()' instead.
##   # Was:
##   data %>% select(col_names_keep)
##
##   # Now:
##   data %>% select(all_of(col_names_keep))
##
## See <https://tidysselect.r-lib.org/reference/faq-external-vector.html>.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```

armed_tidy <- Armed_forces %>%
  filter(!grepl("Total",pay_grade)) %>%
  filter(!grepl("Pay Grade", pay_grade))
armed_tidy <- armed_tidy %>%
  pivot_longer(
    cols = -pay_grade,
    names_to = c("branch", "gender"),
    names_pattern = "[a-z]+_(male|female)",
    values_to = "count"
  )
view(armed_tidy)

#Seperate out rows for frequency table
armed_tidy <- tibble(
  branch = c("Army", "Navy", "Marines", "Air Force", "Space Force"),
  pay_grade = c("E1", "E2", "E3", "E4", "E5"),
  n = c(100, 120, 80, 60, 40)
)
armed_tidy_indiv <- uncount(armed_tidy,n)
view(armed_tidy_indiv)
#Create Frequency Table
table(armed_tidy_indiv)

```

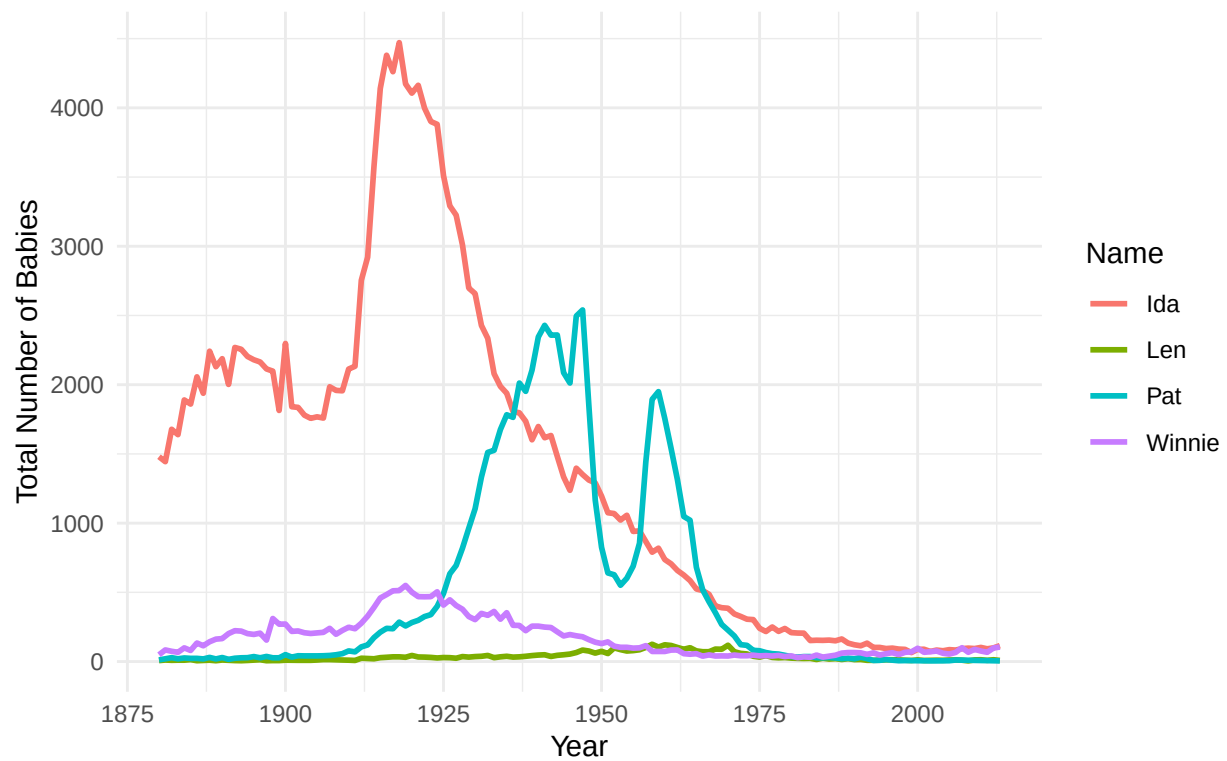
```

##           pay_grade
## branch      E1  E2  E3  E4  E5
## Air Force      0   0   0  60   0
## Army          100   0   0   0   0
## Marines        0   0  80   0   0
## Navy           0 120   0   0   0
## Space Force    0   0   0   0  40

```

#Baby Names Popularity I chose these names because they seemed unique and I wanted 2 girls names and 2 boys names.

Popularity of Baby Names Over Time

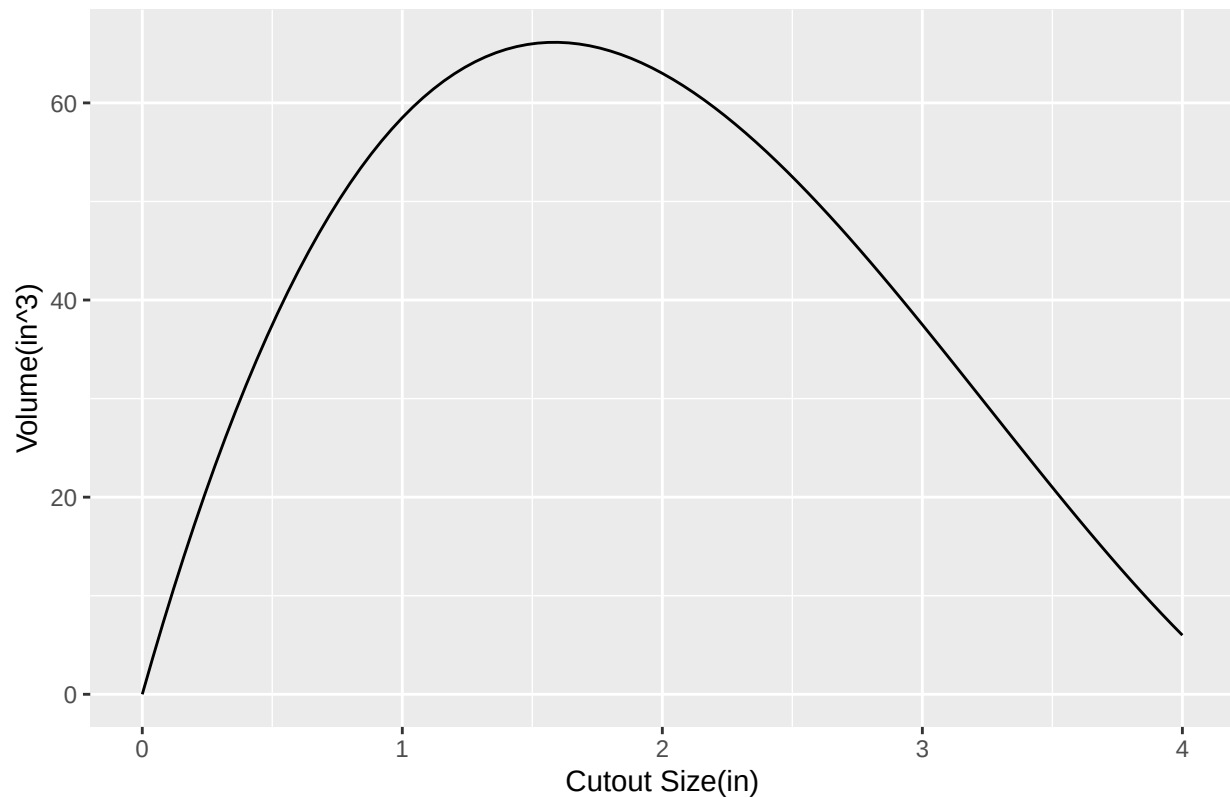


A line chart showing the popularity of four baby names over time beginning in 1880

#Box problem mathematical function

```
#Define volume func for box
V <- function(x) {
  return(x*(11-2*x)*(8.5-2*x))
}
#Plot
ggplot(data.frame(x=c(0,4)),aes(x=x)) +
  stat_function(fun=V) +
  labs(title = "Box Volume vs Cutout Size", x = "Cutout Size(in)",y="Volume(in^3)")
```

Box Volume vs Cutout Size



#What I've Learned so Far

In this course I've learned how to use specific functions in R and new ways to wrangle data to best fit my needs.

#Appendix all code

```
library(tidyr)
library(dplyr)
library(tibble)

#Wrangle Data
col_names <- c("pay_grade", "army_male", "army_female", "army_total", "navy_male", "navy_female", "navy_total")
Armed_forces <- read.csv("US_Armed_Forces_(6_2025) - Sheet1.csv")
colnames(Armed_forces) <- col_names
Armed_forces <- na.omit(Armed_forces)
col_names_keep <- c("pay_grade", "army_male", "army_female", "navy_male", "navy_female", "marinecorps_male")

Armed_forces <- Armed_forces %>%
  select(col_names_keep)
armed_tidy <- Armed_forces %>%
  filter(!grepl("Total", pay_grade)) %>%
  filter(!grepl("Pay Grade", pay_grade))
armed_tidy <- armed_tidy %>%
  pivot_longer(
    cols = -pay_grade,
```

```

    names_to = c("branch", "gender"),
    names_pattern = "([a-z]+)_(male|female)",
    values_to = "count"
  )
view(armed_tidy)

#Seperate out rows for frequency table
armed_tidy <- tibble(
  branch = c("Army", "Navy", "Marines", "Air Force", "Space Force"),
  pay_grade = c("E1", "E2", "E3", "E4", "E5"),
  n = c(100, 120, 80, 60, 40)
)
armed_tidy_indiv <- uncount(armed_tidy,n)
view(armed_tidy_indiv)
#Create Frequency Table
table(armed_tidy_indiv)

```

```

##           pay_grade
## branch      E1  E2  E3  E4  E5
## Air Force      0   0   0  60   0
## Army          100  0   0   0   0
## Marines        0   0  80   0   0
## Navy           0 120   0   0   0
## Space Force    0   0   0   0  40

```

```

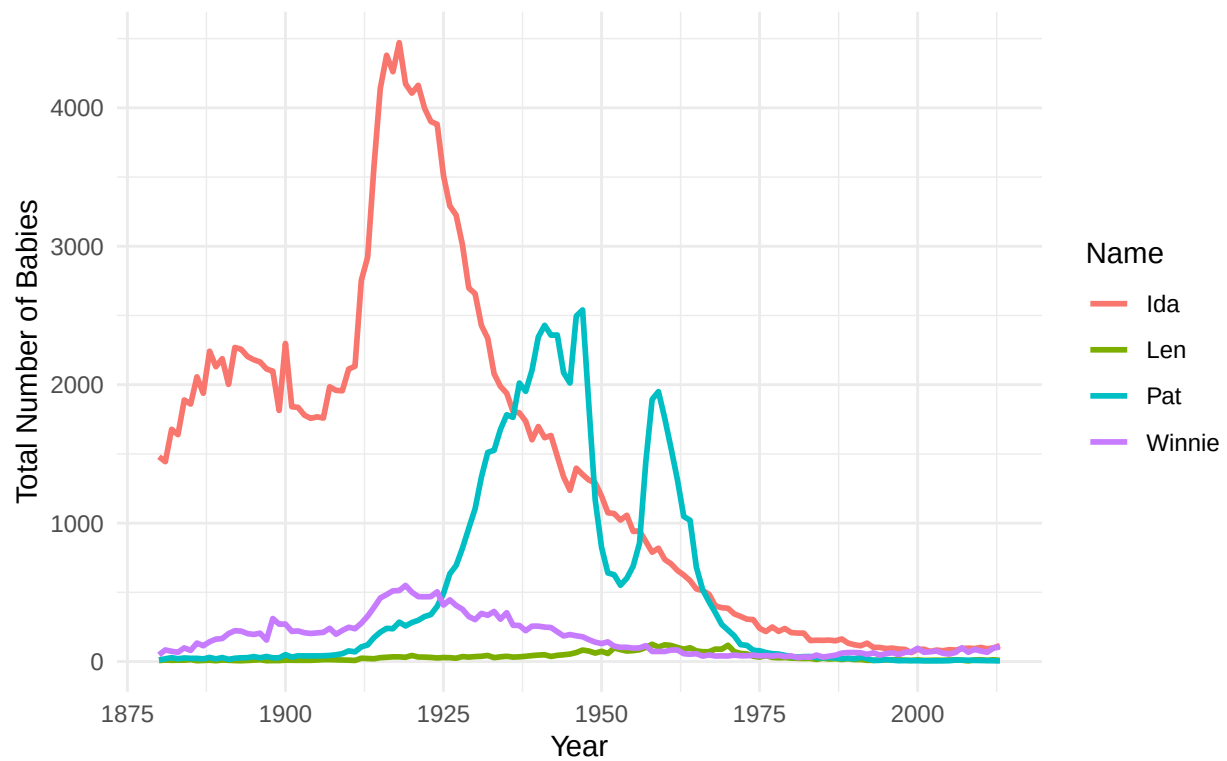
library(dcData)
library(ggplot2)
data("BabyNames")
view(BabyNames)

#Filter BabyNames Data for names to keep
names_to_keep <- c("Winnie", "Ida", "Len", "Pat")
filter_babyNames <- BabyNames %>%
  filter(name %in% names_to_keep) %>%
  group_by(name, year) %>%
  summarise(total=sum(count), .groups = "drop")

#Create Plot to show baby name trends for chosen baby names
ggplot(filter_babyNames, aes(x=year, y=total, color=name)) +
  geom_line(linewidth=1) +
  labs(title = "Popularity of Baby Names Over Time",
       x="Year", y="Total Number of Babies", color = "Name",
       alt = "Line chart showing popularity of baby names over time",
       caption = "A line chart showing the popularity of four baby names over time beginning in 1880") +
  theme_minimal()

```

Popularity of Baby Names Over Time



A line chart showing the popularity of four baby names over time beginning in 1880

```
#Define volume func for box
V <- function(x) {
  return(x*(11-2*x)*(8.5-2*x))
}
#Plot
ggplot(data.frame(x=c(0,4)),aes(x=x)) +
  stat_function(fun=V) +
  labs(title = "Box Volume vs Cutout Size", x = "Cutout Size(in)",y="Volume(in^3)")
```

